PFM-STM32G474 — V4 Design & Functionality

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July 29, 2026

Abstract

Complete pin assignment for revision V4 of the PFM controller board, built on the STM32G474QET6 (LQFP128). Assignments are grouped by peripheral function; the appendix lists all 128 pins in package order. Alternate-function assignments were verified against the STM32G474xB/xC/xE datasheet (DS12288 Rev. 6, Table 12 "Pin definition" and Table 13 "Alternate function") and the STM32CubeMX device database for this package.

Revision	Name	Date	Modification
0.x	Everett Penne	July 29, 2026	Creation

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1 Overview

Item	Value
Device	STM32G474QET6
Package	LQFP128 (14 mm × 14 mm, 0.4 mm pitch)
Core	Arm Cortex-M4F @ 170 MHz
Board revision	V4
Reference document	DS12288 Rev. 6 (STM32G474xB/xC/xE datasheet)

The V4 revision extends the controller from three phases to six (twelve HRTIM outputs with complements), adds six timer-capture inputs for pulse-frequency-modulated current feedback, adds an external QUAD-SPI memory, relocates the general-purpose output bank to port G, and adds a dedicated hardware fault input and a CAN interface. Section 6 tabulates every difference from V3.

Design of this board was initiated due to the following: necessity of ordering more instances of the (V3) controller, correcting footprint and routing mistakes on (V3) of the controller, and new requirements – additional HRTIM outputs (V3 has 3, Transrex needs 4) – from the Transrex supply control upgrade. To the greatest extent possible, pin mappings and conjectured software architecture were preserved from work already completed on the V3 revision of the controller.

This board was designed to satisfy the following:

1. **Limiter controller**
   (a) 3 independent fiber optic PWM channels at 100kHz with complementary PWM pair and deadtime insertion
   (b) Per-period control of PWM duty cycle
   (c) 4x gate drive status inputs per channel (fiber optic)
   (d) 1x analog measurement per channel for active feedback mode
2. **HVPS controller**
   (a) 3 fiber optic PFM channels at 19kHz with complementary PFM pair and deadtime insertion
   (b) Per-period control of PFM frequency
   (c) 12x gate driver status inputs (fiber optic)
   (d) 3x analog measurement per channel for active feedback mode
3. **Transrex controller**
   (a) 4 independent fiber optic PFM channels at 50kHz – 100kHz for Transrex demand signal, sweepable frequency as control signal.
   (b) 4 PFM fiber optic inputs, received from Transrex V-to-f converter to measure Transrex output
   (c) Closed-loop control and fast fault monitoring
   (d) 6 Fiber optic inputs per Transrex unit

Each requirement was carefully considered and assigned to an appropriate peripheral as tabulated and discussed in the following sections. Where possible, if additional parallel peripherals were available on otherwise unassigned pins, functionality (such as output channel counts) was expanded to ensure additional headroom on peripherals, which allows for flexible reprogramming for other tasks.

2 Firmware Architecture

This section describes the software that drives the pins tabulated in Section 4: the module layout, the order in which the firmware brings the board up, and the two-stage mechanism by which a commanded frequency/duty profile becomes a table of hardware register values played back in real time.

2.1 Module Layout

The firmware is organized as a small set of single-responsibility modules, each owning one piece of hardware or one concern:

Module	Responsibility
`pfm`	Builds and plays back the table of per-period frequency/duty register values; the core of the firmware.
`hrtim`	Owns the HRTIM1 peripheral: one-time waveform configuration, starting/stopping the phase outputs, and the low-level register write that applies one table entry.
`state_machine`	The four-state Idle/Armed/Firing/Fault machine (Section 3) that gates when the table above is allowed to play.
`adc`	Analog sense-channel acquisition.
`fault`, `fault_pins`	Fault-condition tracking and the generic, table-driven gate-driver/status-pin scanner.
`gpio`	Static GPIO bank configuration (general-purpose inputs/outputs) not owned by a peripheral-specific module.

Each module exposes a small `_Init()` entry point and otherwise communicates only through explicit function calls. There is no shared global state between modules beyond what is documented at each interface.

2.2 Startup Sequence

Module initialization runs once, in a fixed order, before the main loop ever begins. The order is significant: several modules depend on a peripheral or another module having already reached a known state.

1. **Core clock and tick setup.** The HAL is initialized, the system clock is configured to the device's full core frequency, and the SysTick interrupt priority is corrected immediately afterward, before any peripheral that could contend with it is brought up.

2. **Peripheral hardware configuration.** The GPIO alternate- function mapping, the high-resolution timer peripheral, and the host serial interface are configured, in that order. The timer peripheral must be configured before any module that references it (below); its outputs are configured but not yet started.

3. **Application modules, in dependency order.** General- purpose I/O banks, fault-pin tracking, analog sense-channel acquisition, and the table engine itself are each brought to a known, quiescent state. None of this step drives an output: every module here is initialized to "ready, but idle."

4. **State machine initialization.** Only once every module above has reached its quiescent state does the state machine initialize into IDLE. The state in which the board always boots, and from which the operator must explicitly progress before any output is driven (Section 3).

5. **Enable the periodic handler.** The per-period cycle interrupt is enabled at boot, before the first output is ever fired. This is deliberate: the handler that advances the table index must already be wired up and ready *before* the first shot, not enabled reactively at fire time. Until an output stage is actually started, the handler runs as a no-op.

The result of this ordering is a simple invariant: nothing capable of driving an output pin is touched until every supporting module (fault tracking, sense acquisition, the table engine) is already in a known, safe state.

2.3 The Table: Two Stages, Build Then Play

The firmware's central data structure is a table of per-period register values, with one entry per switching period, played back one entry at a time by the timing hardware. Getting a commanded frequency/duty profile into that table is a deliberate two-stage process: a *track* is described in the abstract (frequency in Hz, duty in %), and is only converted into hardware register counts when the table is built.

2.3.1 Stage 1 — Describing a Track

A track is a sequence of straight-line segments in either the frequency domain or the duty-cycle domain, described independently of hardware units:

- An *init* call establishes the track's starting value: a starting frequency in Hz, or a starting duty in %.
- Any number of *segment* calls follow, each naming the next target value together with a step count and a per-step dwell (how many table entries each interpolated point occupies). A segment linearly ramps from wherever the track currently stands to the named target, landing exactly on the target on its final step.
- A track with no segments at all (init only, or no calls made for that axis) degenerates to a flat run at a single constant value.

Interpolation is computed by scaling the full span by the fractional position of each step (multiplying before dividing, in a wide intermediate type) rather than pre-computing a fixed per-step increment. A naively pre-computed increment silently rounds to zero, and the whole segment collapses to a flat line followed by a single jump on the last step, whenever the total span is smaller than the step count, resulting in a failure mode for the small, fine-grained ramps this mechanism aims to support. Frequency and duty each have their own independent track buffer, so a frequency sweep and a duty sweep might be described with entirely different step counts and shapes for the same shot.

2.3.2 Stage 2 — Building the Table

Building the table is a separate, explicit step that merges whatever tracks exist into the final playback array:

1. **Length resolution.** If one or more axes has a described track, the table's length is that track's length; if multiple axes were tracked, their lengths must agree exactly, or the build is rejected outright. If *no* axis was tracked at all, the table falls back to a flat run at whatever constant frequency/duty values are currently set, sized to fill the configured pulse duration.

2. **Per-entry conversion.** For every table index, the build step reads that index's frequency (from the frequency track, or the flat startup value) and duty (from the duty track(s), or the flat startup value) and converts each from engineering units into the register counts the timing hardware actually consumes: a commanded frequency becomes a period count (the number of timer clock ticks in one switching period), and a commanded duty becomes a compare count (the point within that period at which the output transitions), clamped to hardware-safe bounds on both ends.

3. **Failure is explicit, never silent.** A build that fails (mismatched track lengths, or a profile that would need more entries than the table can hold) clears any previous result rather than leaving a stale table in place. A table is only ever considered valid immediately after a build that reported success.

This two-stage split is what makes the mechanism reusable across very different profile shapes: Stage 1 only ever deals in frequency-Hz and duty-%, with no knowledge of timer register widths or clamping rules; Stage 2 is the one place those hardware details live, and it runs identically regardless of how simple or elaborate the track that fed it was.

2.3.3 Playback

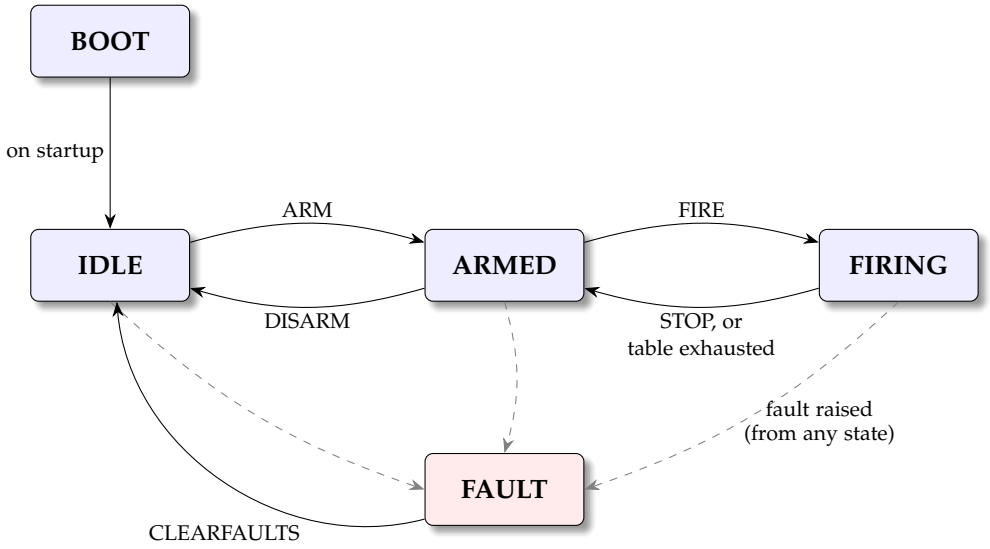
Once built, the table is played back one entry per switching period. Each period, the current entry's register values are written to the timing hardware (skipped entirely if they are identical to what is already loaded, since a long run of unchanging entries is the common case), and a dwell counter determines when to advance to the next index. Reaching the end of the table stops the output cleanly at that boundary and returns control to the state machine (Section 3); nothing about playback requires the table to have been produced by a track sweep as opposed to a flat run. Both are the same array of entries as far as playback is concerned.

2.4 Design Note: Build-Time vs. Play-Time Cost

All of the arithmetic described in Section 2.3: segment interpolation, unit conversion, bounds clamping happens once, at build time, while the output is idle. The play-time path touches only a pre-computed array lookup and a small number of register writes per period, deliberately kept as inexpensive as possible since it runs on the critical timing path. Any future extension to the track/segment mechanism should preserve this split: added complexity belongs in the build step, not the per-period playback step.

3 Controller State Machine

A brief note on how the firmware operates the board, since several pin choices (the FIRE trigger, the hardware fault input, the gate-driver-status inputs) exist to serve specific states. The firmware runs a four-state machine. A pulse is never emitted except from the FIRING state, and every path into FIRING passes through an explicit, guarded ARMED step:



Idle Powered and configured, outputs inactive. Mode and setpoints (frequency, duty, table segments) may be changed only here.

Armed A runnable table has been built and validated; the board is ready to fire but no output is being driven yet.

Firing The HRTIM outputs are live and playing the table. A shot ends when the table is exhausted (automatic return to ARMED), on an operator `STOP`, or on a fault.

Fault Reachable from *any* state. The outputs are forced to their safe level — for the hardware fault input (`HRTIM1_FLT6`, §4.2 below) this happens in silicon with no CPU involvement. `CLEARFAULTS` returns the machine to IDLE, but only once the underlying fault condition (lockout) has cleared.

`ARM` is rejected if the controller is locked out or has no runnable configuration; `FIRE` is accepted only from ARMED. Both the serial `FIRE` command and the hardware fiber-optic trigger reach FIRING through this same guarded path.

4 Pin Mapping by Peripheral

4.1 HRTIM1 — Phase Outputs

Six phases (U, V, W, X, Y, Z), each with a complementary output driven by the HRTIM dead-time unit. All twelve outputs come from HRTIM1 timers A–F on alternate function AF13. See 7.1.

Pin	Port	Signal name (V4)	Peripheral / AF	DS pg.
89	PA8	PHASE_U_PIN	HRTIM1_CHA1	p66
90	PA9	PHASE_UN_PIN	HRTIM1_CHA2	p66
91	PA10	PHASE_V_PIN	HRTIM1_CHB1	p66
92	PA11	PHASE_VN_PIN	HRTIM1_CHB2	p66
66	PB12	PHASE_W_PIN	HRTIM1_CHC1	p63
67	PB13	PHASE_WN_PIN	HRTIM1_CHC2	p63
68	PB14	PHASE_X_PIN	HRTIM1_CHD1	p63
69	PB15	PHASE_XN_PIN	HRTIM1_CHD2	p64
87	PC8	PHASE_Y_PIN	HRTIM1_CHE1	p65
88	PC9	PHASE_YN_PIN	HRTIM1_CHE2	p65
80	PC6	PHASE_Z_PIN	HRTIM1_CHF1	p64
81	PC7	PHASE_ZN_PIN	HRTIM1_CHF2	p65

4.2 HRTIM1 — Hardware Fault Input

Dedicated hardware fault input. A fault event forces the enabled HRTIM outputs to their programmed safe state through the HRTIM output crossbar, with no CPU involvement. See 8.1.3

Pin	Port	Signal name (V4)	Peripheral / AF	DS pg.
100	PC10	HR_FAULT_INPUT_PIN	HRTIM1_FLT6	p68

4.3 Timer Capture — PFM Feedback Inputs

Six pulse-frequency-modulated feedback inputs, measured by general-purpose timer input-capture channels. Each pin is dual-labelled: it serves either as a PFM capture input *or* as the listed general-purpose input, not both simultaneously.

Pin	Port	Signal name (V4)	Timer / AF	DS pg.
99	PA15	PFM_Input_01/GPInput_01	TIM2_CH1	p67
114	PD4	PFM_Input_02/GPInput_05	TIM2_CH2	p69
41	PB2	PFM_Input_03/GPInput_09	TIM5_CH1	p61
102	PC12	PFM_Input_04/GPInput_02	TIM5_CH2	p68
119	PB4	PFM_Input_05/GPInput_06	TIM3_CH1	p70
74	PD12	PFM_Input_06/GPInput_10	TIM4_CH1	p64

4.4 ADC — Analog Inputs

See 8.1.4.

Pin	Port	Signal name (V4)	Peripheral / AF	Secondary	DS pg.
27	PA0	AnalogIn_01	ADC1_IN1	COMP1_INM	p59
28	PA1	AnalogIn_02	ADC1_IN2	COMP1_INP	p59
22	PC0	AnalogIn_03	ADC1_IN6	COMP3_INM	p59
39	PB0	AnalogIn_04	ADC1_IN15	COMP4_INP	p61
23	PC1	AnalogIn_05	ADC1_IN7	COMP3_INP	p59
40	PB1	AnalogIn_08	ADC1_IN12	COMP1_INP	p61

4.5 DAC — Analog Outputs

See 8.1.4.

Pin	Port	Signal name (V4)	Peripheral / AF	Secondary	DS pg.
33	PA4	AnalogOut_01	DAC1_OUT1	ADC2_IN17	p60
34	PA5	AnalogOut_02	DAC1_OUT2	ADC2_IN13	p60

4.6 QUAD-SPI — External Memory

Bank 1 of the QUADSPI interface, for the external NOR flash.

Pin	Port	Signal name (V4)	DS pg.
62	PB10	QUADSPI_CLK	p63
65	PB11	QUADSPI_BK1_NCS	p63
58	PE12	QUADSPI_BK1_IO0	p62
59	PE13	QUADSPI_BK1_IO1	p62
60	PE14	QUADSPI_BK1_IO2	p62
61	PE15	QUADSPI_BK1_IO3	p63

4.7 Serial Communications

Pin	Port	Signal name (V4)	DS pg.
37	PC4	USART1_TX	p60
38	PC5	USART1_RX	p61
29	PA2	USART2_TX	—
32	PA3	USART2_RX	—
70	PD8	USART3_TX	p64
71	PD9	USART3_RX	p64
120	PB5	FDCAN2_RX	p70
121	PB6	FDCAN2_TX	p71

4.8 Gate Driver Status Inputs

Twelve gate-driver fault/status inputs, all on port E, read as a single 16-bit port snapshot. See 8.1.3

Pin	Port	Signal name (V4)	Peripheral / AF
125	PE0	GateDriverStatus_01	GPI
126	PE1	GateDriverStatus_02	GPI
1	PE2	GateDriverStatus_03	GPI
2	PE3	GateDriverStatus_04	GPI
3	PE4	GateDriverStatus_05	GPI
4	PE5	GateDriverStatus_06	GPI
5	PE6	GateDriverStatus_07	GPI
53	PE7	GateDriverStatus_08	GPI
54	PE8	GateDriverStatus_09	GPI
55	PE9	GateDriverStatus_10	GPI
56	PE10	GateDriverStatus_11	GPI
57	PE11	GateDriverStatus_12	GPI

4.9 General-Purpose Inputs

Dedicated general-purpose inputs. Note that GPInput_01, GPInput_02, GPInput_05, GPInput_06, GPInput_09 and GPInput_10 are *not* listed here — they are alternate roles of the PFM capture pins above.

Pin	Port	Signal name (V4)	Peripheral / AF
11	PF4	GPInput_03	GPI
14	PF5	GPInput_04	GPI
16	PF8	GPInput_07	GPI
17	PF9	GPInput_08	GPI
49	PF12	GPInput_11	GPI
50	PF13	GPInput_12	GPI
52	PF15	Fiber_Enable	GPI

4.10 General-Purpose Outputs

The twelve-output bank now lives on port G (plus PD0/PD1). The three enable signals on PC13–PC15 are supplied through the backup-domain power switch and are limited to approximately 3 mA, 2 MHz and 30 pF (DS12288 Rev. 6, Table 12, note 2) — they are suitable for slow enable lines only.

Pin	Port	Signal name (V4)	Peripheral / AF
82	PG0	GPOut_01	GPO
83	PG1	GPOut_02	GPO
84	PG2	GPOut_03	GPO
85	PG3	GPOut_04	GPO
86	PG4	GPOut_05	GPO
103	PG5	GPOut_06	GPO
104	PG6	GPOut_07	GPO
105	PG7	GPOut_08	GPO
106	PG8	GPOut_09	GPO
107	PG9	GPOut_10	GPO
108	PD0	GPOut_11	GPO
109	PD1	GPOut_12	GPO
7	PC13	GPOut_Enable_Pin	GPO
8	PC14	STM_Enable_Pin	GPO
9	PC15	PWM_Alt_Enable	GPO

4.11 Clock, Reset and Debug

Pin	Port	Signal name (V4)	DS pg.
19	PF0	HSE_IN	p58
20	PF1	HSE_OUT	p58
21	PG10	NRST	p58
96	PA13	SWDIO	p67
98	PA14	SWCLK	p67
118	PB3	SWO	p70
123	PB8	BOOT0	p71

4.12 Power and Ground

Pin	Port	Signal name (V4)	DS pg.
6	VBAT	3V3	p57
12	VSS	GND	p58
13	VDD	3V3	p58
30	GND	GND	p60
31	VDD	3V3	p60
42	VSSA	GND	p61
43	VREF+	VREF+ Capacitor	p61
44	VREF-	VREF- Capacitor	p61
45	VDDA	3.3VA	p61
46	GND	GND	p61
47	VDD	3V3	p61
63	GND	GND	p63
64	VDD	3V3	p63
78	GND	GND	p64
79	VDD	3V3	p64
94	GND	GND	p67
95	VDD	VDD	p67
110	GND	GND	p69
111	VDD	3V3	p69
127	GND	GND	p72
128	VDD	3V3	p72

5 Unassigned Pins

Pins with no V4 connection. Several retain capabilities worth reserving as test pads even if unpopulated; the notes column records the most useful alternate function available on each.

Pin	Port	V3 function	Notable available function
10	PF3	GPInput_02	—
15	PF7	GPInput_06	QUADSPI1_BK1_IO2 (alternate)
18	PF10	GPInput_09	QUADSPI1_CLK (alternate)
24	PC2	—	ADC1_IN8 / ADC2_IN8
25	PC3	—	ADC1_IN9 / ADC2_IN9
26	PF2	GPInput_01	—
35	PA6	—	DAC2_OUT1 (third DAC); OPAMP2_VOUT; ADC2_IN3
36	PA7	—	OPAMP2_VINP; ADC2_IN4; COMP2
48	PF11	GPInput_10	—
51	PF14	—	—
72	PD10	GPOut_11	—
73	PD11	GPOut_12	—
75	PD13	—	TIM4_CH2
76	PD14	—	TIM4_CH3; OPAMP2_VINP
77	PD15	—	TIM4_CH4
93	PA12	—	HRTIM1_FLT1 --- second hardware fault input

Pin	Port	V3 function	Notable available function
97	PF6	GPINPUT_05	—
101	PC11	USART3_RX	UART4_RX / USART3_RX
112	PD2	GPOut_03	—
113	PD3	GPOut_04	—
115	PD5	GPOut_06	USART2_TX
116	PD6	GPOut_07	USART2_RX
117	PD7	GPOut_08	—
122	PB7	—	I2C1_SDA; TIM3_CH4 / TIM4_CH2
124	PB9	—	FDCAN1_TX; I2C1_SDA; TIM4_CH4

6 Changes from V3

Every pin whose function differs between V3 and V4.

Pin	Port	V3	V4
10	PF3	GPInput_02	*unassigned*
15	PF7	GPInput_06	*unassigned*
18	PF10	GPInput_09	*unassigned*
23	PC1	No connection	AnalogIn_05
24	PC2	No connection	*unassigned*
25	PC3	No connection	*unassigned*
26	PF2	GPInput_01	*unassigned*
33	PA4	No connection	AnalogOut_01
34	PA5	No connection	AnalogOut_02
35	PA6	No connection	*unassigned*
36	PA7	No connection	*unassigned*
41	PB2	No connection	PFM_Input_03/GPInput_09
43	VREF+	No connection	VREF+ Capacitor
44	VREF-	No connection	VREF- Capacitor
48	PF11	GPInput_10	*unassigned*
51	PF14	No connection	*unassigned*
58	PE12	No connection	QUADSPI_BK1_IO0
59	PE13	No connection	QUADSPI_BK1_IO1
60	PE14	No connection	QUADSPI_BK1_IO2
61	PE15	No connection	QUADSPI_BK1_IO3
62	PB10	HR_FAULT_INPUT_PIN	QUADSPI_CLK
65	PB11	AnalogIn_05	QUADSPI_BK1_NCS
68	PB14	AnalogIn_06	PHASE_X_PIN
69	PB15	AnalogIn_07	PHASE_XN_PIN
70	PD8	GPOut_09	USART3_TX
71	PD9	GPOut_10	USART3_RX
72	PD10	GPOut_11	*unassigned*
73	PD11	GPOut_12	*unassigned*
74	PD12	No connection	PFM_Input_06/GPInput_10
75	PD13	No connection	*unassigned*
76	PD14	No connection	*unassigned*

Pin	Port	V3	V4
77	PD15	No connection	*unassigned*
80	PC6	No connection	PHASE_Z_PIN
81	PC7	No connection	PHASE_ZN_PIN
82	PG0	No connection	GPOut_01
83	PG1	U_Alt_Pin	GPOut_02
84	PG2	UN_Alt_Pin	GPOut_03
85	PG3	V_Alt_Pin	GPOut_04
86	PG4	VN_Alt_Pin	GPOut_05
87	PC8	W_Alt_Pin	PHASE_Y_PIN
88	PC9	WN_Alt_Pin	PHASE_YN_PIN
93	PA12	No connection	*unassigned*
97	PF6	GPINPUT_05	*unassigned*
98	PA14	swclk	SWCLK
99	PA15	No connection	PFM_Input_01/GPInput_01
100	PC10	USART3_TX	HR_FAULT_INPUT_PIN
101	PC11	USART3_RX	*unassigned*
102	PC12	No connection	PFM_Input_04/GPInput_02
103	PG5	No connection	GPOut_06
104	PG6	No connection	GPOut_07
105	PG7	No connection	GPOut_08
106	PG8	No connection	GPOut_09
107	PG9	No connection	GPOut_10
108	PD0	GPOut_01	GPOut_11
109	PD1	GPOut_02	GPOut_12
112	PD2	GPOut_03	*unassigned*
113	PD3	GPOut_04	*unassigned*
114	PD4	GPOut_05	PFM_Input_02/GPInput_05
115	PD5	GPOut_06	*unassigned*
116	PD6	GPOut_07	*unassigned*
117	PD7	GPOut_08	*unassigned*
119	PB4	No connection	PFM_Input_05/GPInput_06
120	PB5	No connection	FDCAN2_RX
121	PB6	No connection	FDCAN2_TX
122	PB7	No connection	*unassigned*
124	PB9	No connection	*unassigned*

7 Design Justifications

7.1 HRTIM1 Phase Outputs

The V4 controller is configurable for up to 6 HRTIM channel output phases (A-F), plus their complement. The complementary outputs are generated through the STM32G4xx's built-in HRTIM hardware, and deadtime insertion is automatic and easily configurable prior to runtime. The 6-phase outputs builds upon the V3's 3-phase output functionality, and requires no updates to underlying software architecture.

7.2 HRTIM1 Fault Input

The pin PC10 is assigned as HR_FAULT_INPUT_PIN. This pin may be used to halt PFM outputs from the controller if it measures a rising or falling edge signal. While this gating of the HRTIM outputs is handled exclusively via hardware, the configuration is done in software. All 12 Gate Driver Status pins are connected to the HRTIM fault pin, and may be configured as either normally HIGH or normally LOW configurations. All 6 channels are wired to unique fiber optic outputs, as are their respective complements.

7.3 PFM Feedback Inputs

6 timer channels are available wired to fiber optic inputs – TIM2_CH1, TIM2_CH2, TIM5_CH1, TIM5_CH2, TIM3_CH1, and TIM4_CH1. When not in use for PFM capture, the respective pins for these timers are configured as general purpose inputs. Fiber optic PFM inputs to the controller are measured using input capture, which captures frequency and duty cycle in hardware by measuring rising and falling edges with timestamps. This avoids software overhead, which would otherwise require 600,000 function calls at 50kHz - 100kHz input range, the approximate quoted PFM output from the Transrex interface boards, for instance (the Transrex control system is the primary application for the PFM input).

All analog channels selected are 32-bit analog sense and output channels.

7.4 Analog Inputs and Outputs

Six analog inputs and 2 analog outputs are included on the V4 board design. Each is technically configurable as either and input or an output, and the output buffer op amps on the board are reconfigurable on hardware to reflect this.

7.5 QUAD-SPI External Memory

A persistent issue on V2 and V3 of the PFM controller is one of limited memory. The STM32G474 (and most other STM32 microcontroller families) has 128KB of SRAM, where much of the firmware is stored along with PFM array tables. On the V3 board, we have space enough to store 3 phases of PFM table for non-feedback modes outputting a shot up to 50ms at 100kHz. In more general terms, for a 3-phase output we are able to define 5000 periods worth of PFM table values (Phase 1-3 frequencies, Phase 1-3 duty cycles – 6 arrays worth of values). This finite table size is not a consideration for closed-loop feedback mode, where an effectively arbitrarily long shot length is possible.

To alleviate this, an experimental "extra credit" design choice was made for the PFM controller V4. This controller includes on board a Winspace W25Q128JVS IC flash memory allowing expansion of memory up to 128MB using the STM32G4xx's QSPI external memory protocol. In theory, the inclusion of this IC allows circular loading of PFM tables from memory, extending the shot length ceiling. Implementation of this feature is not a priority deliverable, but is essentially free to include since all QSPI Bank1 ports are available.

7.6 Serial Communications

Three USART channels are available – USART 1, 2, & 3. Communication with the controller is performed primarily over UART2. U(S)ARTs 1 & 3 are included primarily as a consideration for other future functionality. For instance, both channels 1 & 3 are configurable to transmit and receive over fiber optic inputs and outputs to allow isolated serial communication to controllable devices, like charging supplies.

Additionally, FDCAN2 is also configurable to transmit and receive over fiber optic, although this is a very non-standard (though not useless) way to perform FDCAN2 communication.

7.7 General-Purpose I/O Banks

A major difference between versions V3 and V4 of the PFM controller is the location of the general-purpose IO banks. For V3, the GPO's were located on bank PD. Since on V4 PFM_Input_02 and PFM_Input_06 required pins PD4 and PD12, respectively, the GPO's were moved mostly to bank PG which was unused save for PG10's NRST line. GPOut_01 through GPOut_10 now correspond to PG0 through PG9, while GPOut_11 and GPOut_12 are at PD0 and PD1.

7.8 Reserved and Unassigned Pins

Reserved pins are reserved to protect MCU power and firmware flashing functionality. Unassigned pins are unassigned because no projected functionality could be determined, or due to physical routing constraints on the PCB layout.

8 Board Layout

8.1 Layout & Routing Choices

The PFM V4 controller board is a 6-layer PCB. The enclosure-interior-facing layer is the "back copper" layer. This layer prioritizes local signal traces, for instance from an IC to itself, or an IC to a nearby IC. All local pin access to the MCU chip is handled on this layer, along with decoupling capacitors and crystal oscillator configuration.

Best layout practice typically says that decoupling capacitors be placed beneath respective integrated circuits to achieve minimally inductive power routing. On the PFM V4 controller board, where possible this principle was violated to minimize the amount of in-lab desk assembly time. This choice prioritizes the automatic component soldering at a PCB fabricator, keeping local labor time low.

In some cases, trace routing or other component placements took precedent, and decoupling capacitors or pullup resistors had to be placed on the "front copper" layer, facing the box front panel. These components will need to be hand-assembled.

8.1.1 MCU

Power, crystal oscillator, and debug for the onboard MCU was taken from ST RM0440, DS12288 Rev 6, and Phil's Lab, and has been validated on the V3 version of the PFM controller.

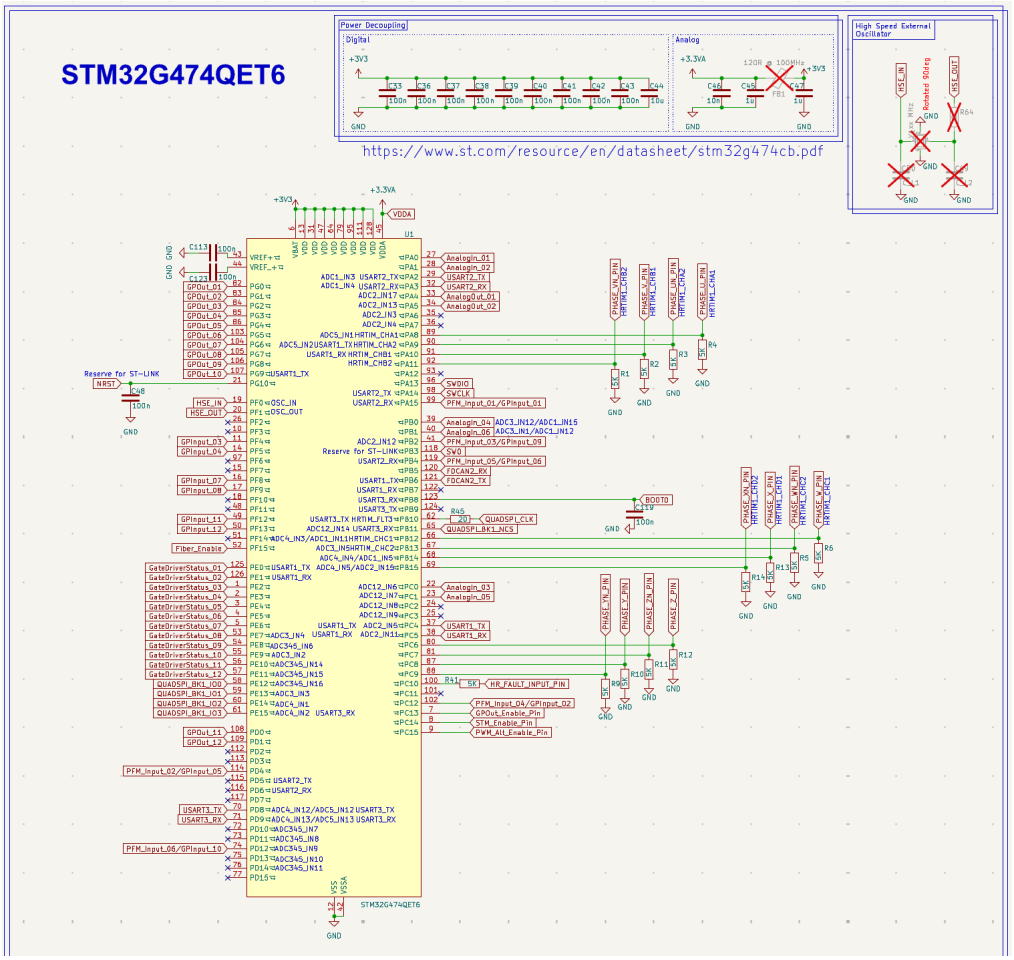


Figure 1: Schematic pin mapping of stm32g474qet6.

8.1.2 PFM Output

The fiber-optic PFM output on the V4 controller may be configured for between 3 and 6 phase output. This is done by physically soldering resistors to select the maximum number of outputs. It is possible that this may be completed solely via software, however this feature requires more testing to validate.

Each timer channel, for instance Phase_U and Phase_UN, corresponding to TIMA_CH1 and TIMA_CH2, respectively, is inverted via an SN74LVC2G14DBVR, before being ANDed with its complement to drive an output Versatile Link transmitter via IX4340NTR gate driver. That is,

$$\texttt{PHASE_OUT} = (\texttt{! PHASE }) \&\& \texttt{PHASE_N}$$

This ensures that each fiber illuminates only when both MCU timer channel outputs are outputting in complementary mode, minimizing the risk of hardware shoot-through on the supply side.

Outputs are gateable via software. To gate an output, the PWM_Enable and/or PWM_Alt_Enable pins must be pulled to low. This feature is bypassable via on-board hardware (swapping pull-up resistors – see KiCAD file for specifics).

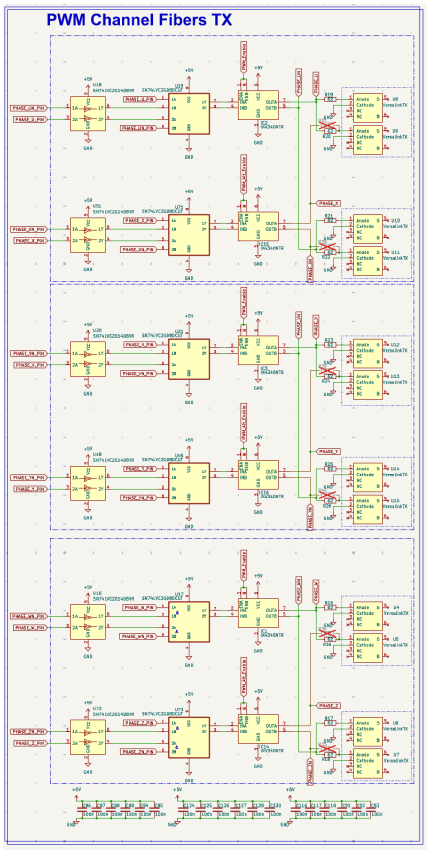


Figure 2: PFM fiber optic output hardware block with cross-shoot protection.

8.1.3 Hardware Fault & Enable Handling

All 12 Gate Driver Status signals are wired directly to Port E, pins PE0 through PE11 (corresponding to GateDriverStatus_01 through GateDriverStatus_12, respectively) and are monitored via software polling. To ensure fast handling of gate driver faults, if necessary, additional onboard hardware

logic separate from the MCU is available for both Normally HIGH and Normally LOW, processed in parallel and selectable via resistor. Available hardware may be configured prior to installation of the controller to select which output enables are gated by a gate driver fault.

The hardware-processed fault signal is additionally routed to pin PC10, a fast HRTIM fault pin which gates HRTIM outputs in the MCU via internal MCU hardware, again bypassing software. Since this fast fault handling collapses all gate driver signals into a single fault signal, information on which gate driver(s) triggered the fault is not available until software polling completes or until the operator directly polls the gate driver status pins.

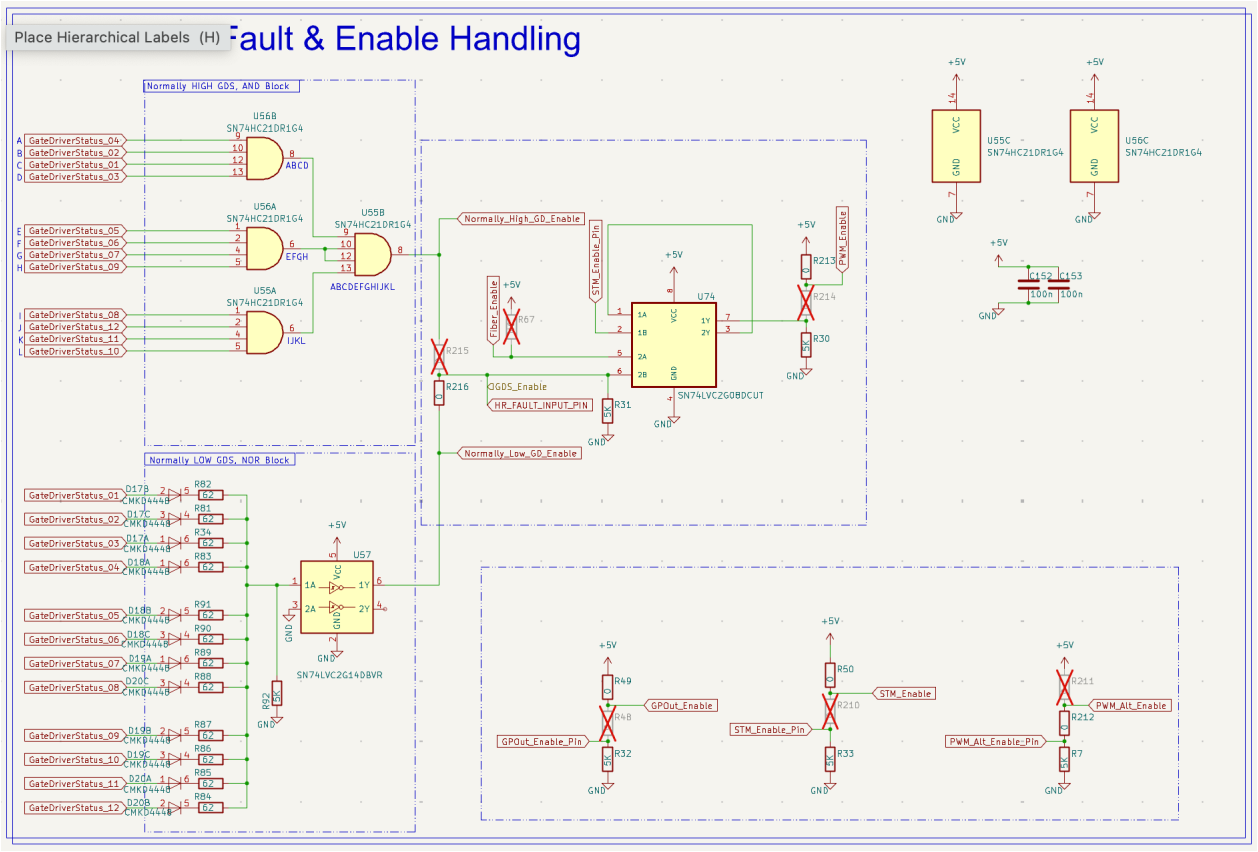


Figure 3: Gate Drive Status processing and fault handling configuration.

8.1.4 Analog Inputs and Outputs

As a default configuration, 6 analog input channels and 2 analog output channels are available for feedback or voltage-level comparator fault inputs. All 8 analog channels may be reconfigured as any combination of analog inputs or outputs, though this requires hardware modifications (the board design and layout makes these modifications available).

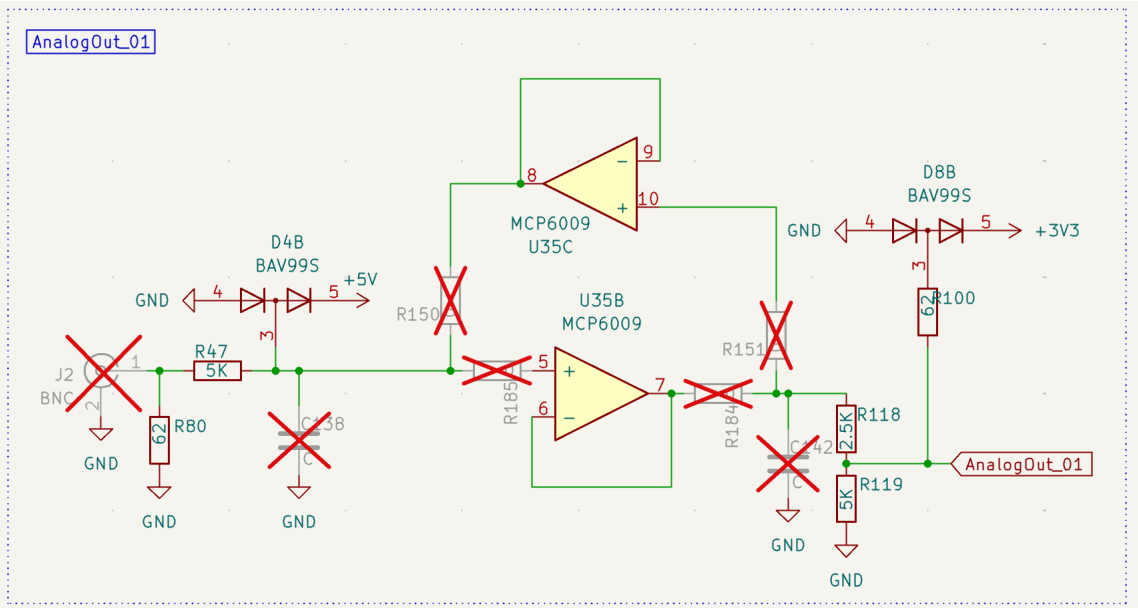


Figure 4: Analog input/output buffer, configurable to match the function of the relevant analog pin.

9 Box Layout

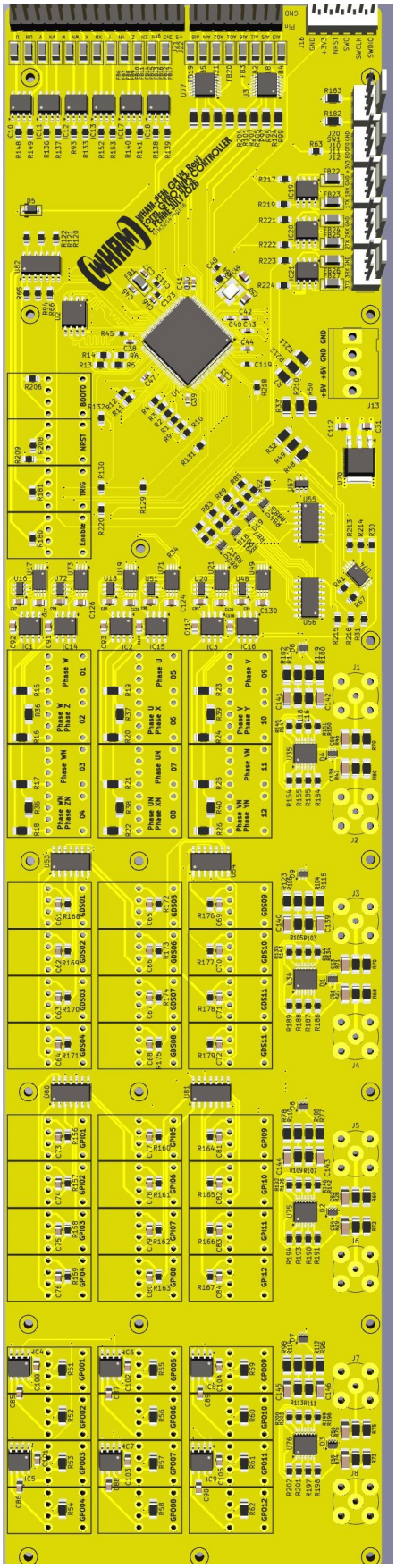


Figure 5: 3D model of PFM V4 controller, facing box interior (away from front panel)

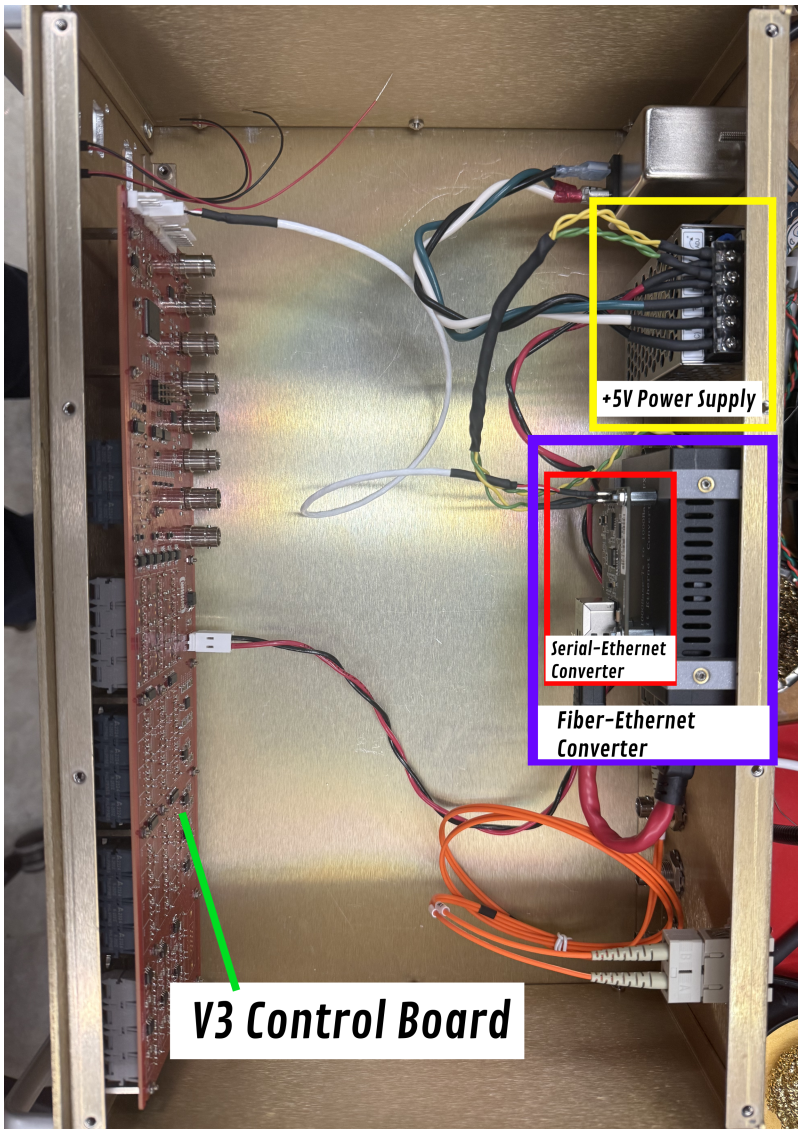


Figure 6: Box interior with labelled components.

A Complete Pin Listing (Package Order)

All 128 pins in package order.

Pin	Port	Signal name (V4)	Peripheral / AF	DS pg.
1	PE2	GateDriverStatus_03	GPI	—
2	PE3	GateDriverStatus_04	GPI	—
3	PE4	GateDriverStatus_05	GPI	—
4	PE5	GateDriverStatus_06	GPI	—
5	PE6	GateDriverStatus_07	GPI	—
6	VBAT	3V3	—	p57
7	PC13	GPOut_Enable_Pin	GPO	—
8	PC14	STM_Enable_Pin	GPO	—
9	PC15	PWM_Alt_Enable	GPO	—
10	PF3	*unassigned*	—	—
11	PF4	GPInput_03	GPI	—
12	VSS	GND	—	p58
13	VDD	3V3	—	p58
14	PF5	GPInput_04	GPI	—
15	PF7	*unassigned*	—	—
16	PF8	GPInput_07	GPI	—
17	PF9	GPInput_08	GPI	—
18	PF10	*unassigned*	—	—
19	PF0	HSE_IN	—	p58
20	PF1	HSE_OUT	—	p58
21	PG10	NRST	—	p58
22	PC0	AnalogIn_03	ADC1_IN6	p59
23	PC1	AnalogIn_05	ADC1_IN7	p59
24	PC2	*unassigned*	—	—
25	PC3	*unassigned*	—	—
26	PF2	*unassigned*	—	—
27	PA0	AnalogIn_01	ADC1_IN1	p59
28	PA1	AnalogIn_02	ADC1_IN2	p59
29	PA2	USART2_TX	USART2_TX	—
30	GND	GND	—	p60
31	VDD	3V3	—	p60
32	PA3	USART2_RX	USART2_RX	—
33	PA4	AnalogOut_01	DAC1_OUT1	p60
34	PA5	AnalogOut_02	DAC1_OUT2	p60
35	PA6	*unassigned*	—	—
36	PA7	*unassigned*	—	—
37	PC4	USART1_TX	USART1_TX	p60
38	PC5	USART1_RX	USART1_RX	p61
39	PB0	AnalogIn_04	ADC1_IN15	p61
40	PB1	AnalogIn_08	ADC1_IN12	p61
41	PB2	PFM_Input_03/GPInput_09	GPI	p61
42	VSSA	GND	—	p61
43	VREF+	VREF+ Capacitor	—	p61

Pin	Port	Signal name (V4)	Peripheral / AF	DS pg.
44	VREF-	VREF- Capacitor	—	p61
45	VDDA	3.3VA	—	p61
46	GND	GND	—	p61
47	VDD	3V3	—	p61
48	PF11	*unassigned*	—	—
49	PF12	GPInput_11	GPI	—
50	PF13	GPInput_12	GPI	—
51	PF14	*unassigned*	—	—
52	PF15	Fiber_Enable	GPI	—
53	PE7	GateDriverStatus_08	GPI	—
54	PE8	GateDriverStatus_09	GPI	—
55	PE9	GateDriverStatus_10	GPI	—
56	PE10	GateDriverStatus_11	GPI	—
57	PE11	GateDriverStatus_12	GPI	—
58	PE12	QUADSPI_BK1_IO0	QUADSPI_BK1_IO0	p62
59	PE13	QUADSPI_BK1_IO1	QUADSPI_BK1_IO1	p62
60	PE14	QUADSPI_BK1_IO2	QUADSPI_BK1_IO2	p62
61	PE15	QUADSPI_BK1_IO3	QUADSPI_BK1_IO3	p63
62	PB10	QUADSPI_CLK	QUADSPI_CLK	p63
63	GND	GND	—	p63
64	VDD	3V3	—	p63
65	PB11	QUADSPI_BK1_NCS	QUADSPI_BK1_NCS	p63
66	PB12	PHASE_W_PIN	HRTIM1_CHC1	p63
67	PB13	PHASE_WN_PIN	HRTIM1_CHC2	p63
68	PB14	PHASE_X_PIN	HRTIM1_CHD1	p63
69	PB15	PHASE_XN_PIN	HRTIM1_CHD2	p64
70	PD8	USART3_TX	USART3_TX	p64
71	PD9	USART3_RX	USART3_RX	p64
72	PD10	*unassigned*	—	—
73	PD11	*unassigned*	—	—
74	PD12	PFM_Input_06/GPInput_10	GPI	p64
75	PD13	*unassigned*	—	—
76	PD14	*unassigned*	—	—
77	PD15	*unassigned*	—	—
78	GND	GND	—	p64
79	VDD	3V3	—	p64
80	PC6	PHASE_Z_PIN	HRTIM1_CHF1	p64
81	PC7	PHASE_ZN_PIN	HRTIM1_CHF2	p65
82	PG0	GPOut_01	GPO	—
83	PG1	GPOut_02	GPO	—
84	PG2	GPOut_03	GPO	—
85	PG3	GPOut_04	GPO	—
86	PG4	GPOut_05	GPO	—
87	PC8	PHASE_Y_PIN	HRTIM1_CHE1	p65
88	PC9	PHASE_YN_PIN	HRTIM1_CHE2	p65
89	PA8	PHASE_U_PIN	HRTIM1_CHA1	p66
90	PA9	PHASE_UN_PIN	HRTIM1_CHA2	p66

Pin	Port	Signal name (V4)	Peripheral / AF	DS pg.
91	PA10	PHASE_V_PIN	HRTIM1_CHB1	p66
92	PA11	PHASE_VN_PIN	HRTIM1_CHB2	p66
93	PA12	*unassigned*	—	—
94	GND	GND	—	p67
95	VDD	VDD	—	p67
96	PA13	SWDIO	—	p67
97	PF6	*unassigned*	—	—
98	PA14	SWCLK	—	p67
99	PA15	PFM_Input_01/GPInput_01	GPI	p67
100	PC10	HR_FAULT_INPUT_PIN	HRTIM1_FLT6	p68
101	PC11	*unassigned*	—	—
102	PC12	PFM_Input_04/GPInput_02	GPI	p68
103	PG5	GPOut_06	GPO	—
104	PG6	GPOut_07	GPO	—
105	PG7	GPOut_08	GPO	—
106	PG8	GPOut_09	GPO	—
107	PG9	GPOut_10	GPO	—
108	PD0	GPOut_11	GPO	—
109	PD1	GPOut_12	GPO	—
110	GND	GND	—	p69
111	VDD	3V3	—	p69
112	PD2	*unassigned*	—	—
113	PD3	*unassigned*	—	—
114	PD4	PFM_Input_02/GPInput_05	GPI	p69
115	PD5	*unassigned*	—	—
116	PD6	*unassigned*	—	—
117	PD7	*unassigned*	—	—
118	PB3	SWO	—	p70
119	PB4	PFM_Input_05/GPInput_06	GPI	p70
120	PB5	FDCAN2_RX	GPI	p70
121	PB6	FDCAN2_TX	GPO	p71
122	PB7	*unassigned*	—	—
123	PB8	BOOT0	—	p71
124	PB9	*unassigned*	—	—
125	PE0	GateDriverStatus_01	GPI	—
126	PE1	GateDriverStatus_02	GPI	—
127	GND	GND	—	p72
128	VDD	3V3	—	p72

B Board Layers

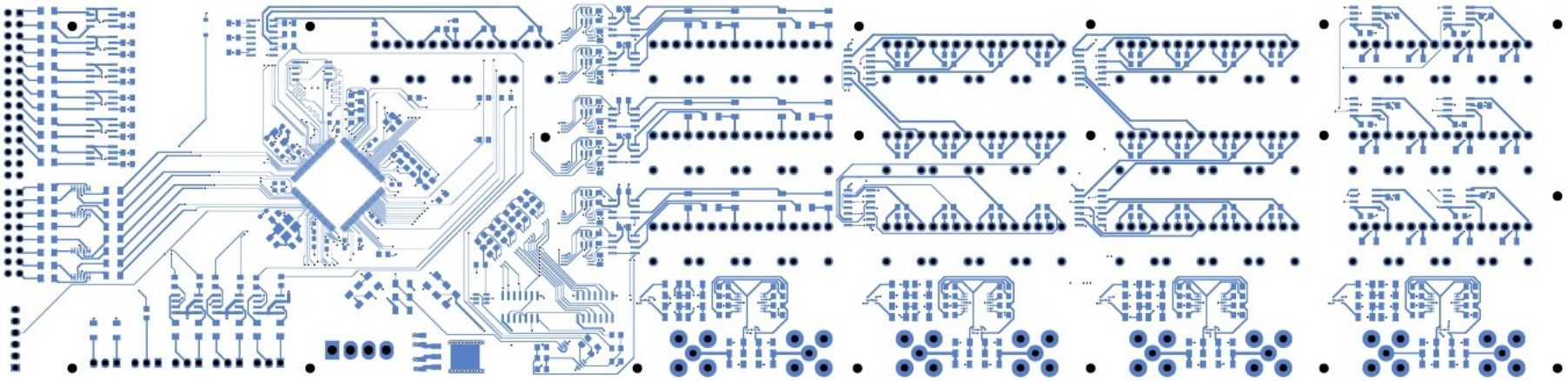


Figure 7: Back copper layer. Priority routing given to local signal traces. Long distance traces used where conflicts arose with other layers' routing.

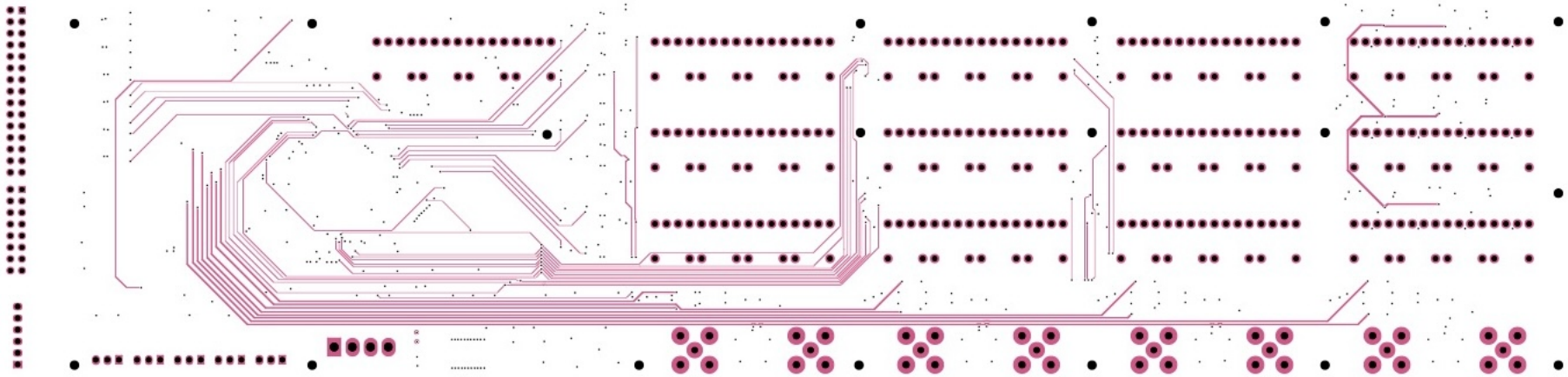


Figure 8: Inner copper layer 4. Priority routing given to HRTIM traces and analog traces.

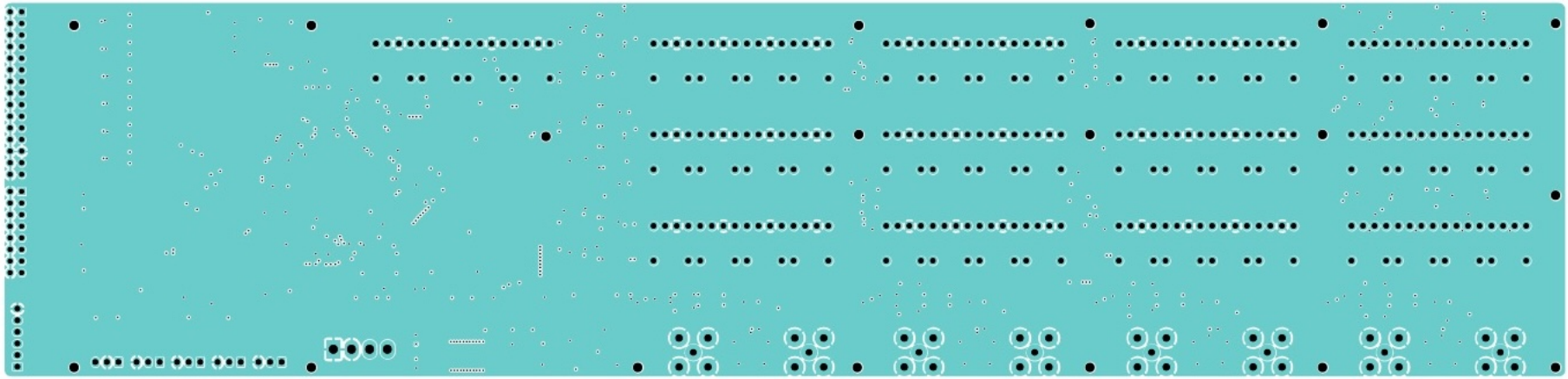


Figure 9: Inner copper layer 3. Unified GND plane. No trace routing allowed.

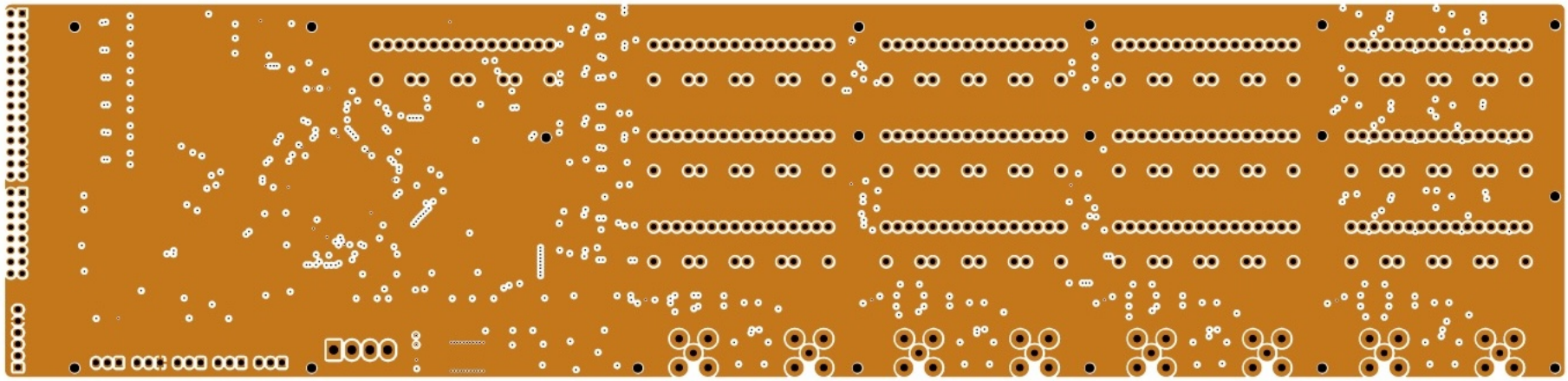


Figure 10: Inner copper layer 2. Unified +3V3 power plane. No trace routing allowed.

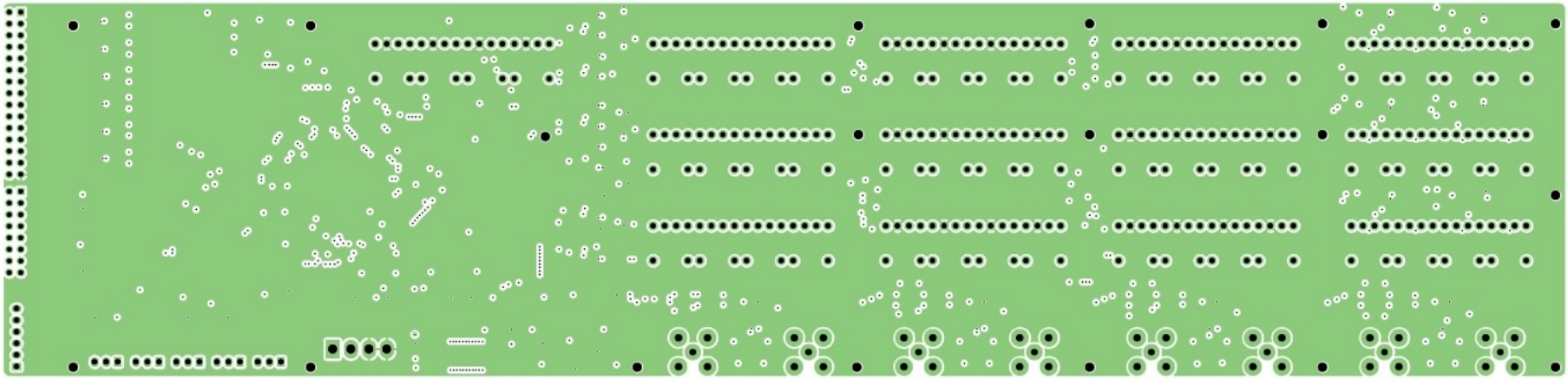


Figure 11: Inner copper layer 1. Unified +5V power plane. No trace routing allowed.

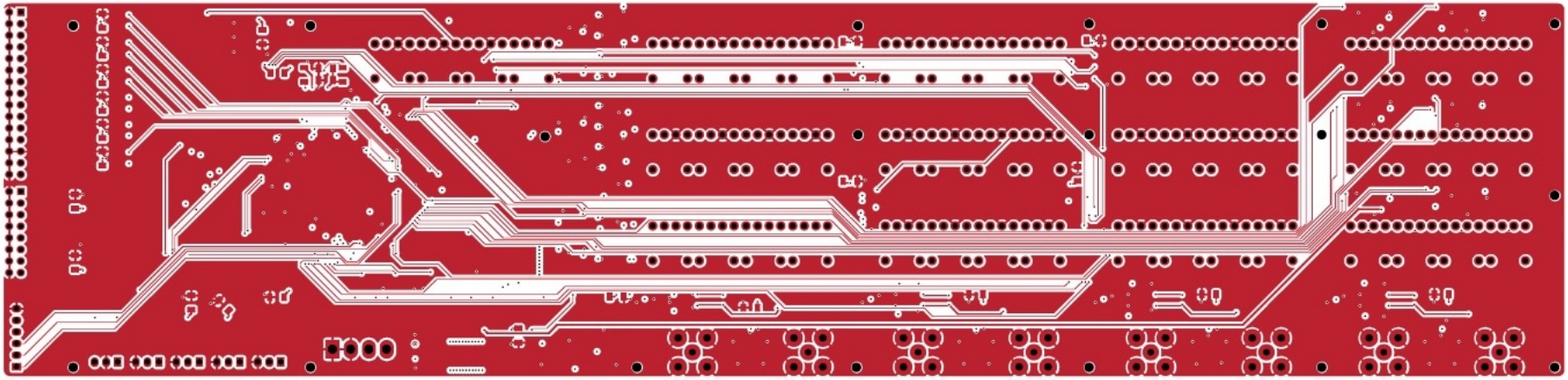


Figure 12: Front copper layer. General-purpose digital signal routing plane with GND fill. Slow signal traces preferred to route on this plane, including general purpose inputs and outputs, as well as diagnostic output traces.

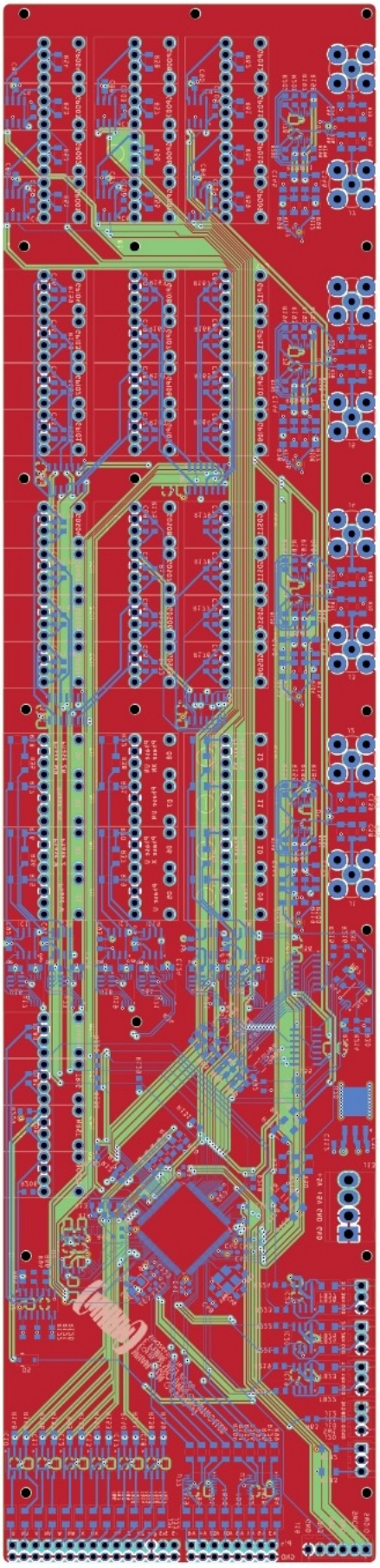


Figure 13: Full PCB stackup.